

GLUCare

An Advanced Non-Invasive
Glucose Monitoring System
using NIR Spectroscopy

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Associate Professor

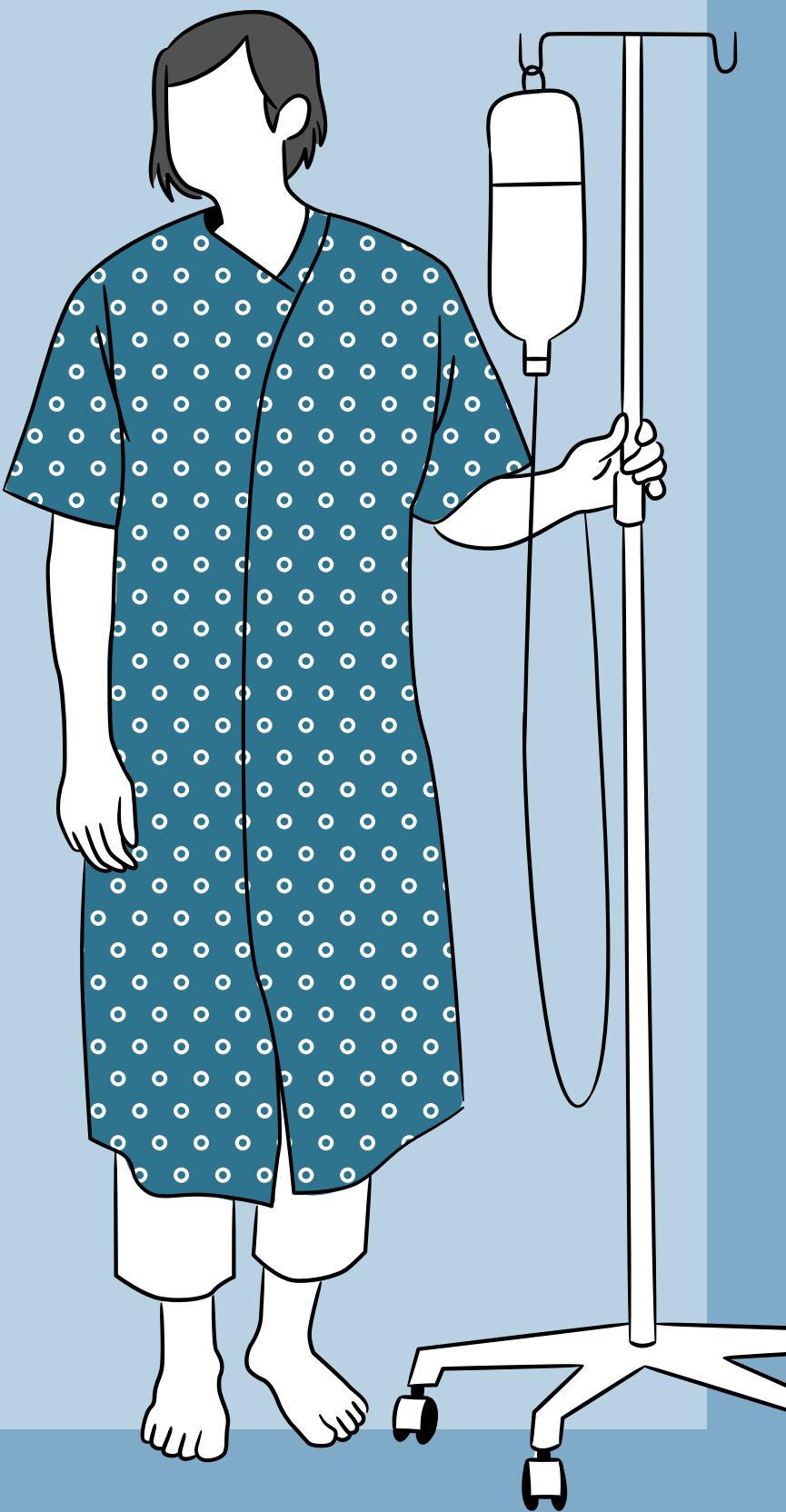
CPG No. 10





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Background of Project

Non-invasive glucose measurement using Near-Infrared (NIR) spectroscopy offers a promising alternative to traditional methods, but commercialization has been hindered by accuracy and precision challenges.

This project aims to address these issues by integrating clinical factors that account for physiological differences, improving the accuracy and reliability of NIR-based detection, and paving the way for successful commercialization.



Project Scope and Utility

The scope of this project is to develop an accurate, reliable, and affordable non-invasive glucose monitoring sensor that is easy to use and compatible with existing diabetes management systems.

Additionally , we will create a website where patients and doctors can log in to track glucose levels.

Project Objectives

- 1.** To analyze and compare existing state of the art technologies for glucose monitoring.
- 2.** To design and implement a non-invasive device to monitor Blood Glucose level using Machine Learning.
- 3.** To create the test data and validate the performance of the device by comparing with clinical invasive techniques.
- 4.** To design and develop a web application/app to display the results.

Literature Survey

Research Paper

Tools/Technology

Findings

Non-Invasive Blood Glucose Monitoring Technology

NIRS, Electrochemical Sensors, Ultrasound, Thermal Sensors, Multispectral and Hyperspectral Imaging

NIR spectroscopy :non-invasive alternative to traditional blood glucose testing. Technological Advancements: Real-time glucose measurements without the need for blood samples, improving patient compliance and ease of use.

Non-invasive methods of glucose measurement: current status and future perspectives

Support vector regression (SVR), Principal component regression (PCR) , and Artificial neural network (ANN)

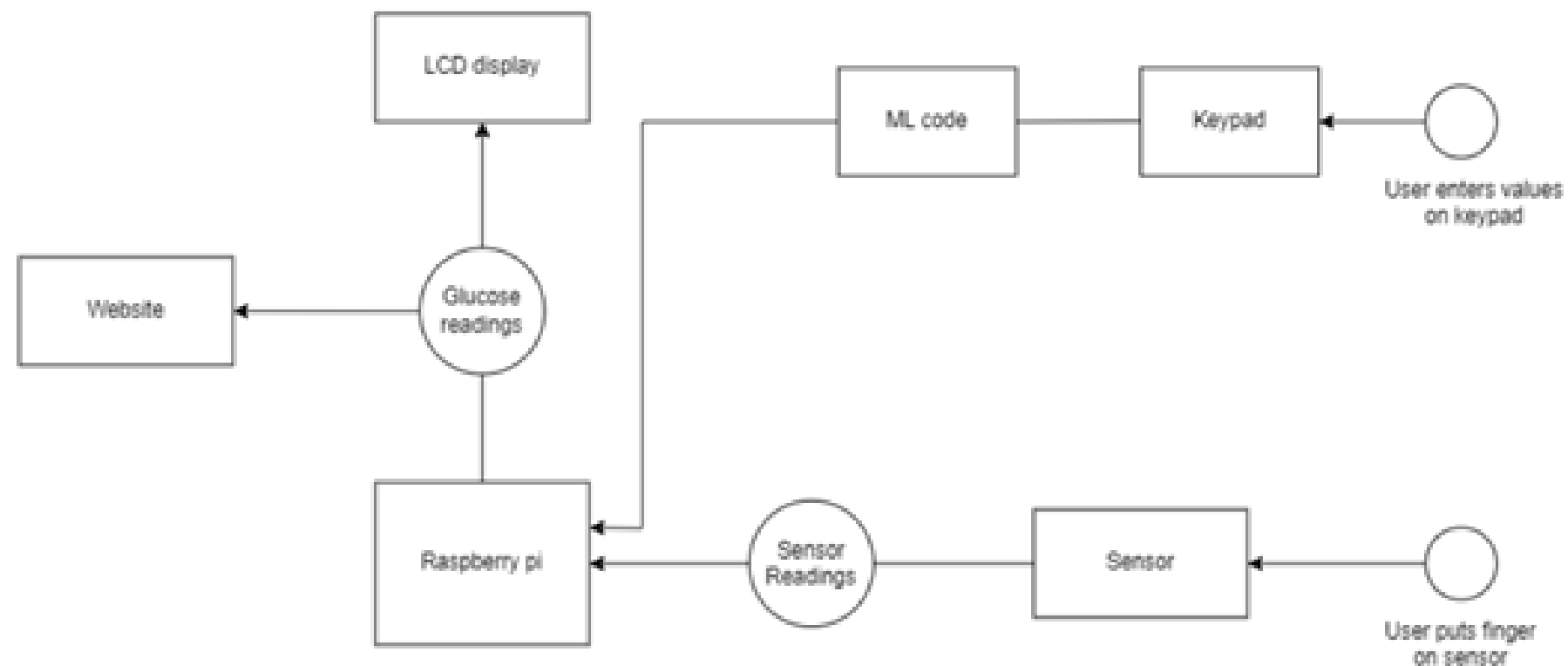
This paper examines non-invasive blood glucose monitoring as a means of improving diabetes care. It discusses the implications of tested devices and highlights the shortcomings of existing approaches.

Wearable-band type visible-near infrared optical biosensor for non-invasive blood glucose monitoring

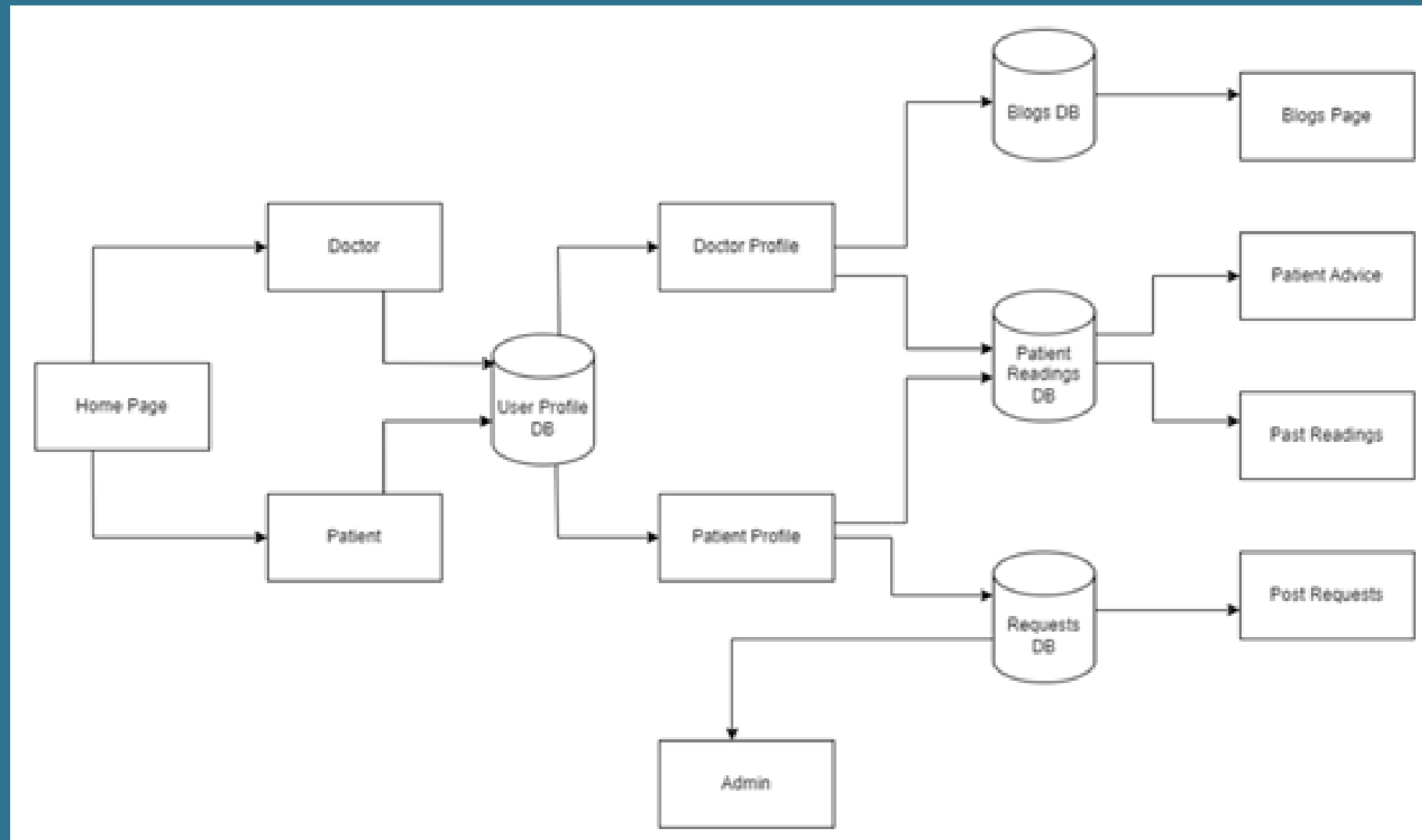
NIR Sensor, Digital wavelet transform

A wearable blood glucose sensor using visible-near infrared spectroscopy shows promise with a correlation coefficient of 0.86 and a prediction error of 6.16 mg/dl in trials with 12 volunteers.

Architecture Design

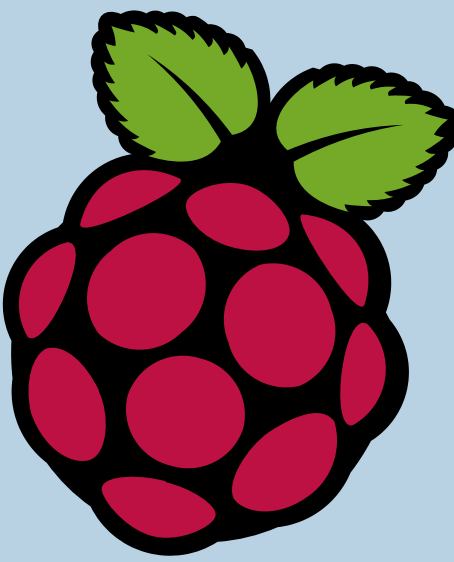


Component Diagram



Website Architecture

Techniques & Tools Used



TOOLS USED

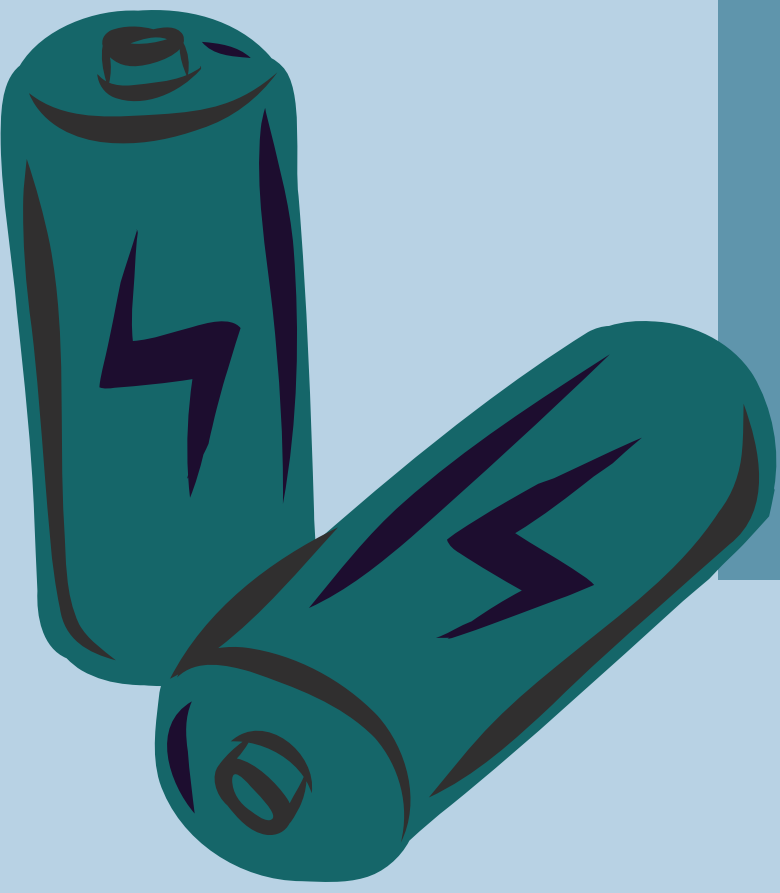
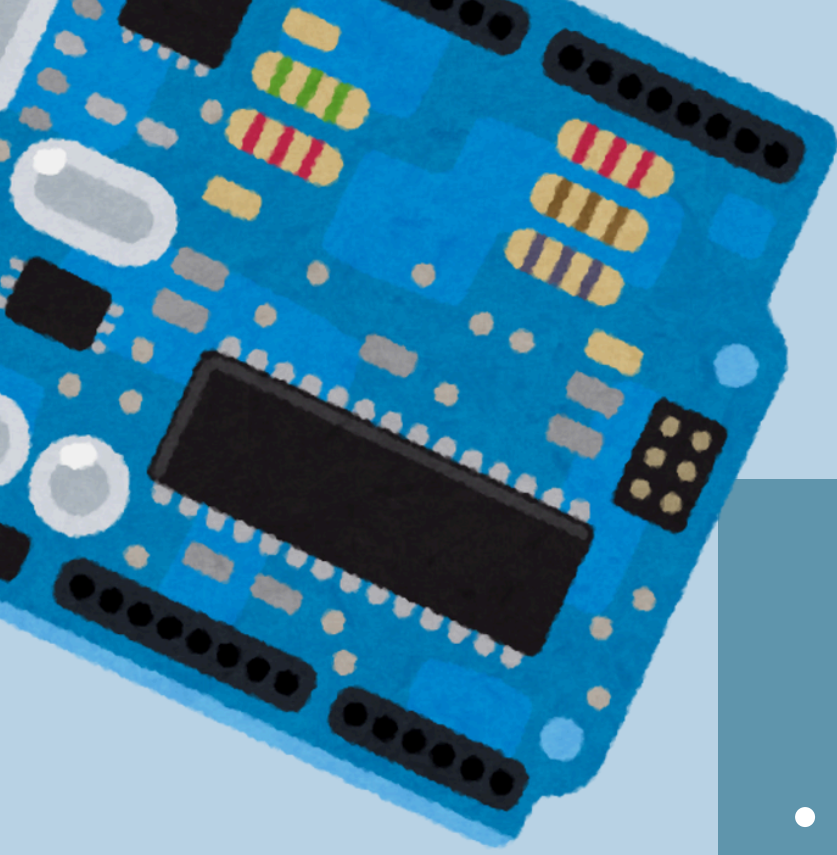
- NIR Sensor(TCRT5000)
- Arduino UNO
- Raspberry pi
- Wifi Board ESP8266
- LCD
- Buzzer and Button
- Lithium Battery
- Connectors
- Cables

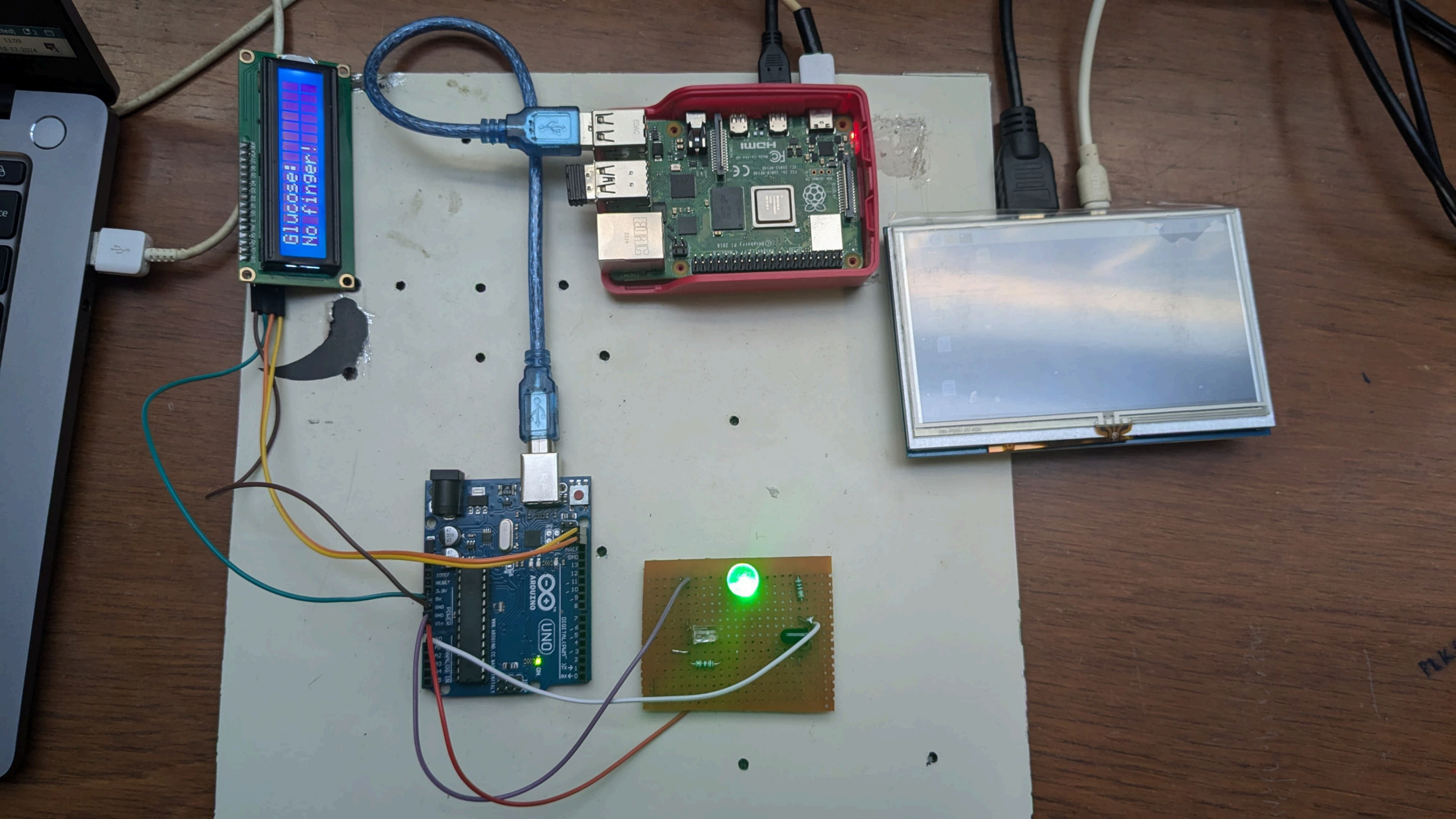
PLATFORMS

- Wokwi Simulator
- Tinkercad
- Arduino IDE

TECH-STACK

- **React.js**
- Firebase
- MongoDB





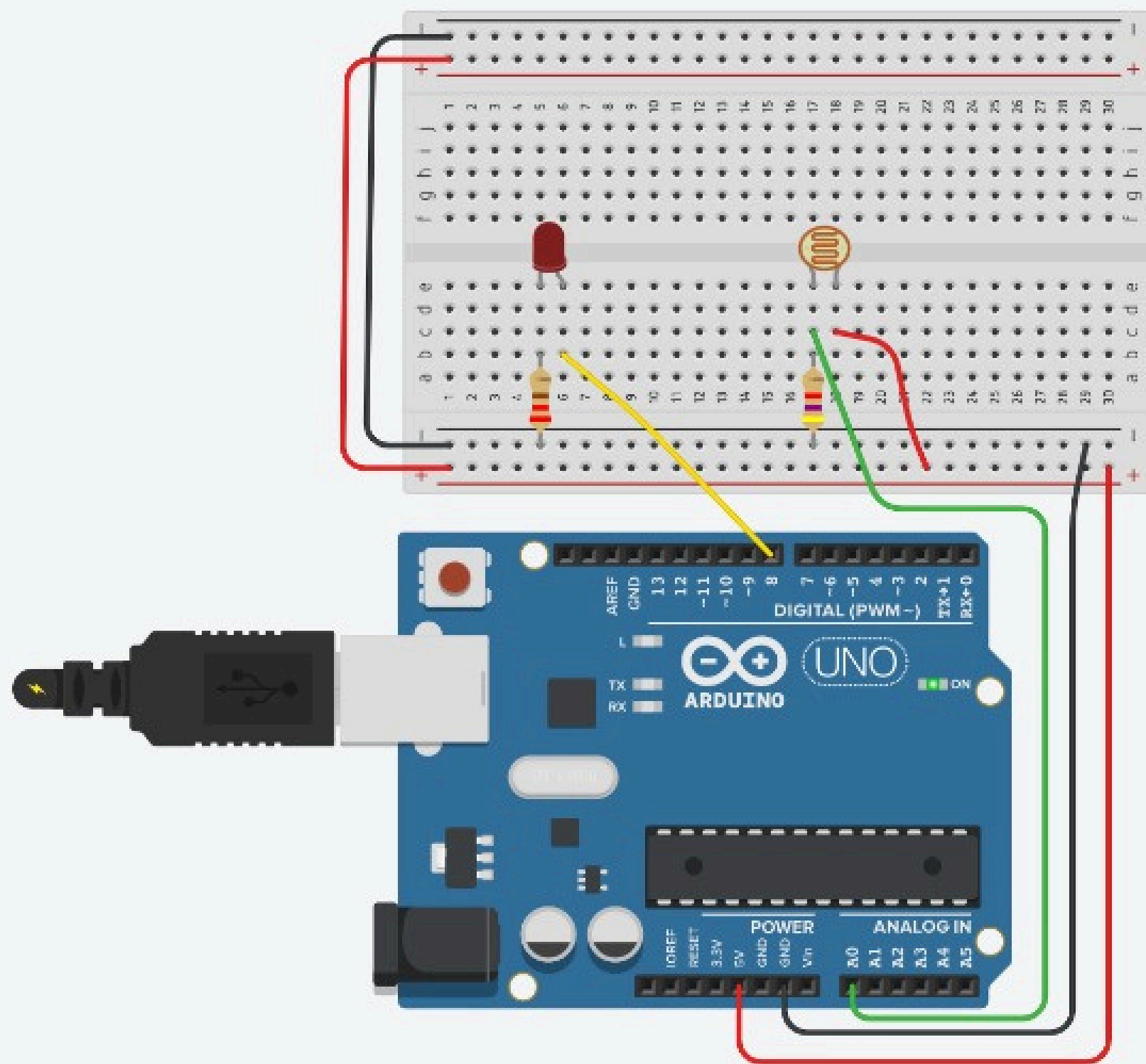
tedl 2
13:09
18-12-2024

Glucose: 100 mg/dL
No finger!

ARDUINO
UNO
DIGITAL COUNTER

1

5 inch LCD



```
1  int ledPin = 8;
2  int ldrPin = A0;
3  int ldrValue = 0;
4
5  void setup() {
6    pinMode(ledPin, OUTPUT);
7    Serial.begin(9600);
8  }
9
10 void loop() {
11   ldrValue = analogRead(ldrPin);
12   Serial.println(ldrValue);
13
14   // Calculate the glucose level using the regression formula
15   int glucoseLevel = ((0.000009) * (ldrValue) * (ldrValue)) + (0.
16
17   // Display the glucose level in the Serial Monitor
18   Serial.print("Glucose Level: ");
19   Serial.println(glucoseLevel);
20
21   // Use the glucose level to control the LED
22   if (glucoseLevel > 120) { // Example threshold value for high
23     digitalWrite(ledPin, HIGH);
24   } else {
25     digitalWrite(ledPin, LOW);
26   }
27 }
```

Serial Monitor

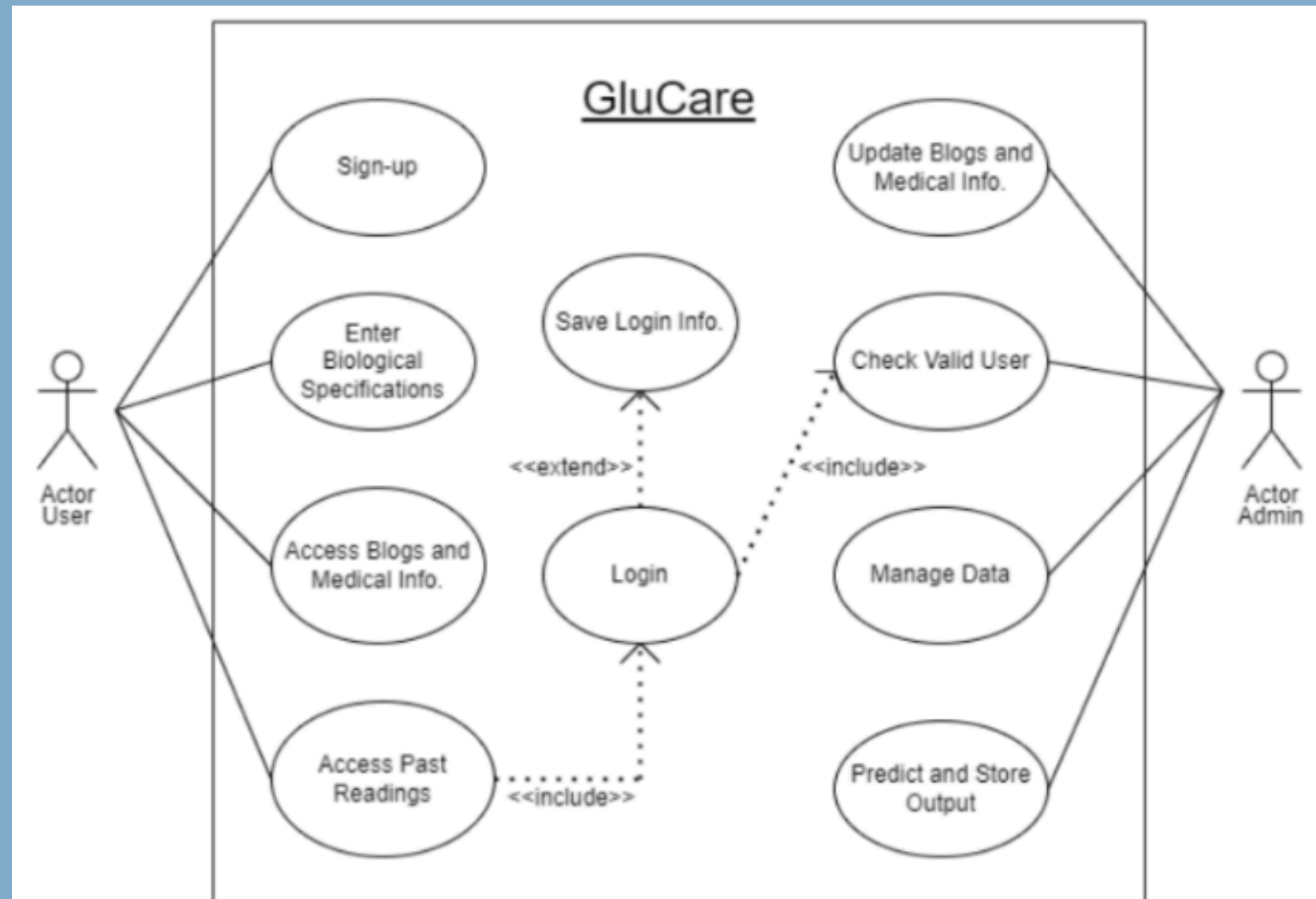
```
389
Glucose Level: 120
389
Glucose Level: 120
389
Glucose Level: 120
```

Send

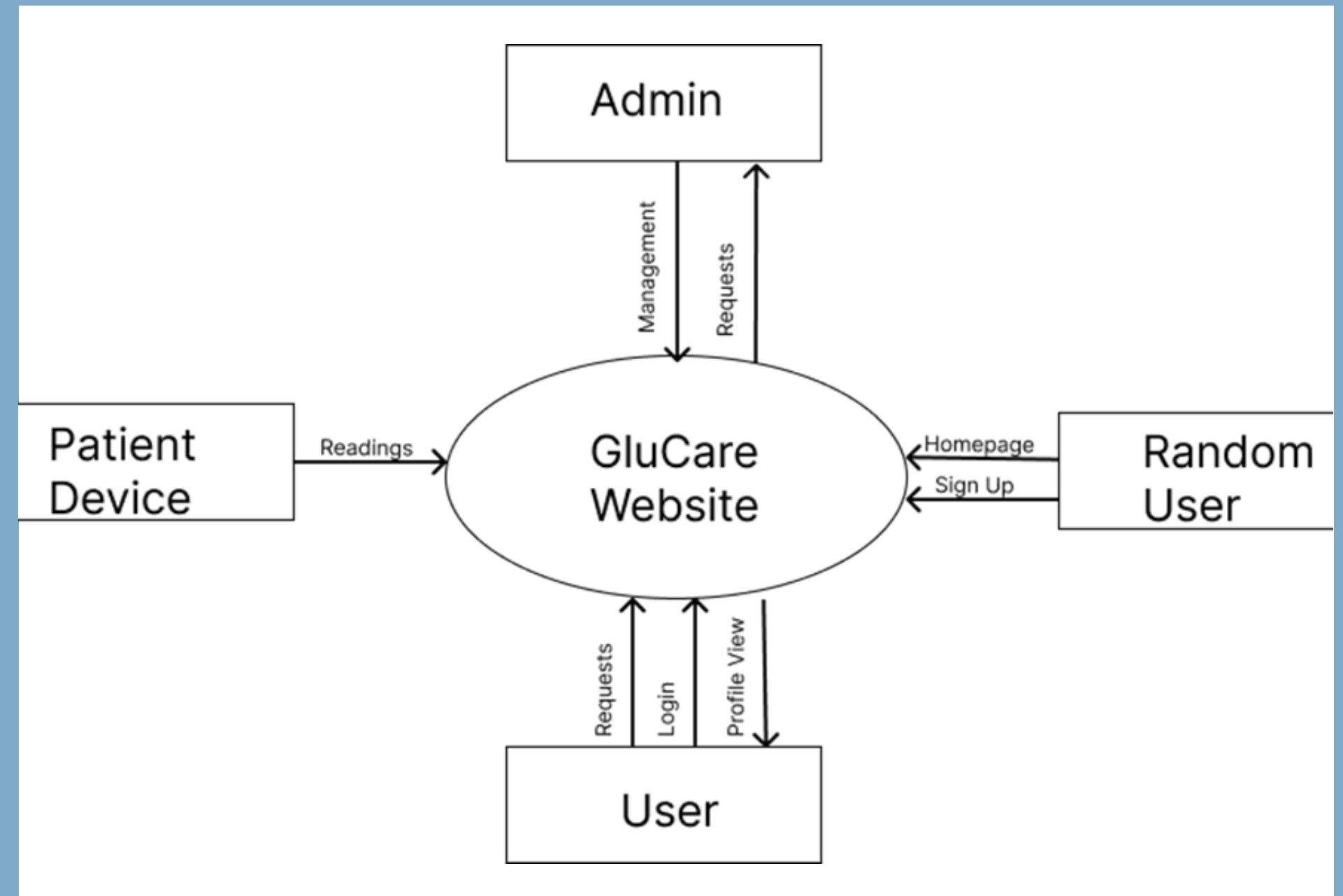
Clear



UML DIAGRAMS

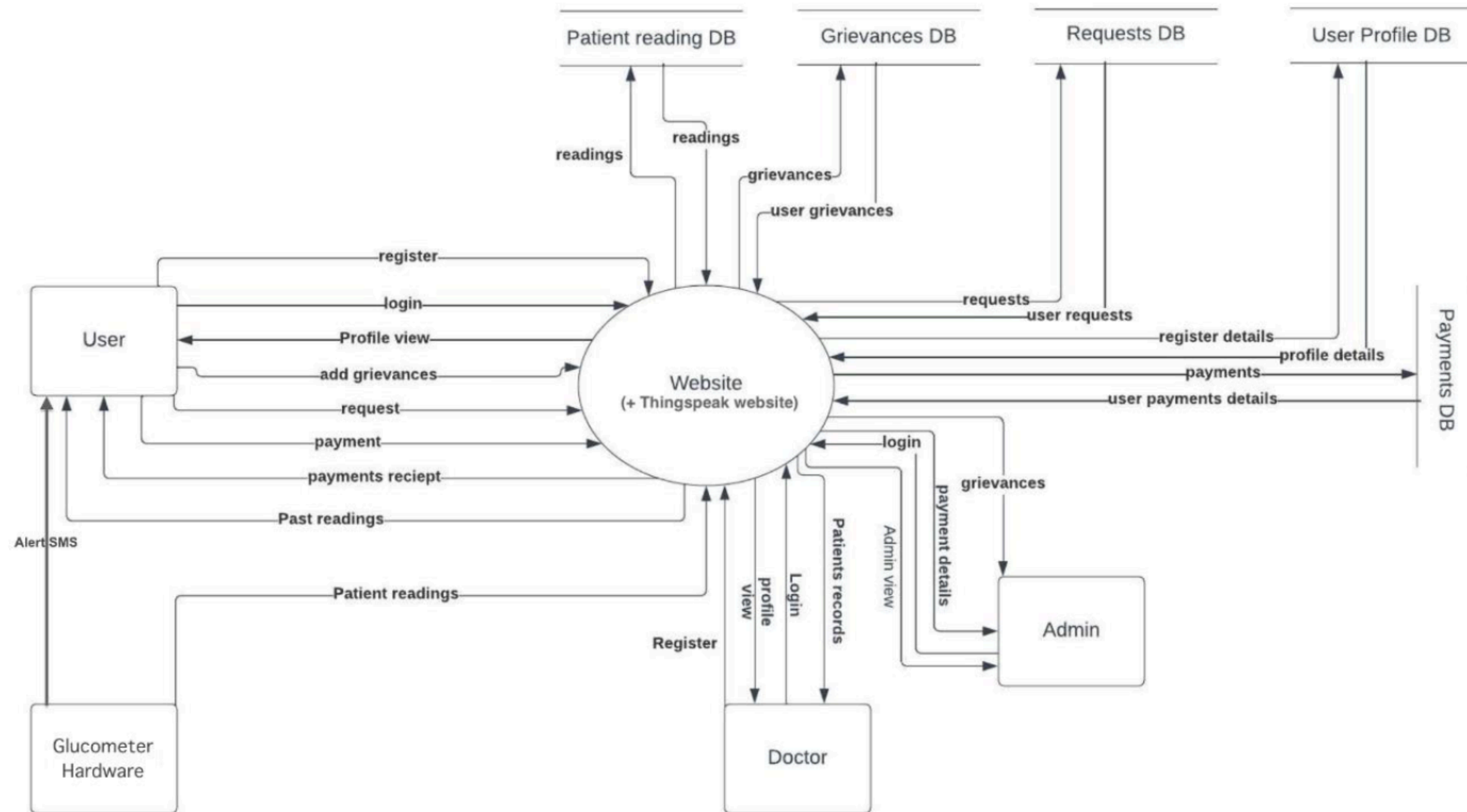


USE CASE DIAGRAM

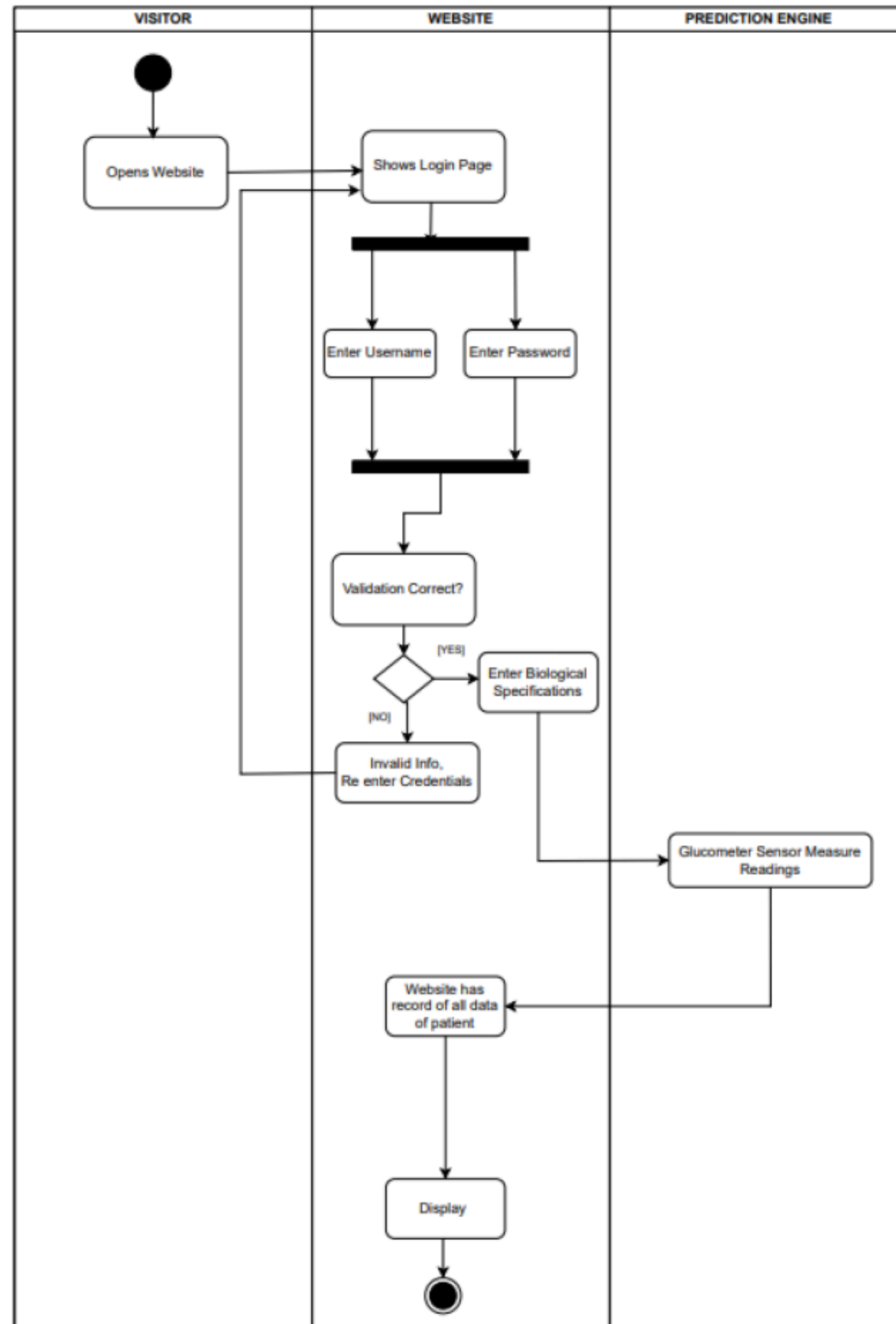


DFD LEVEL 0

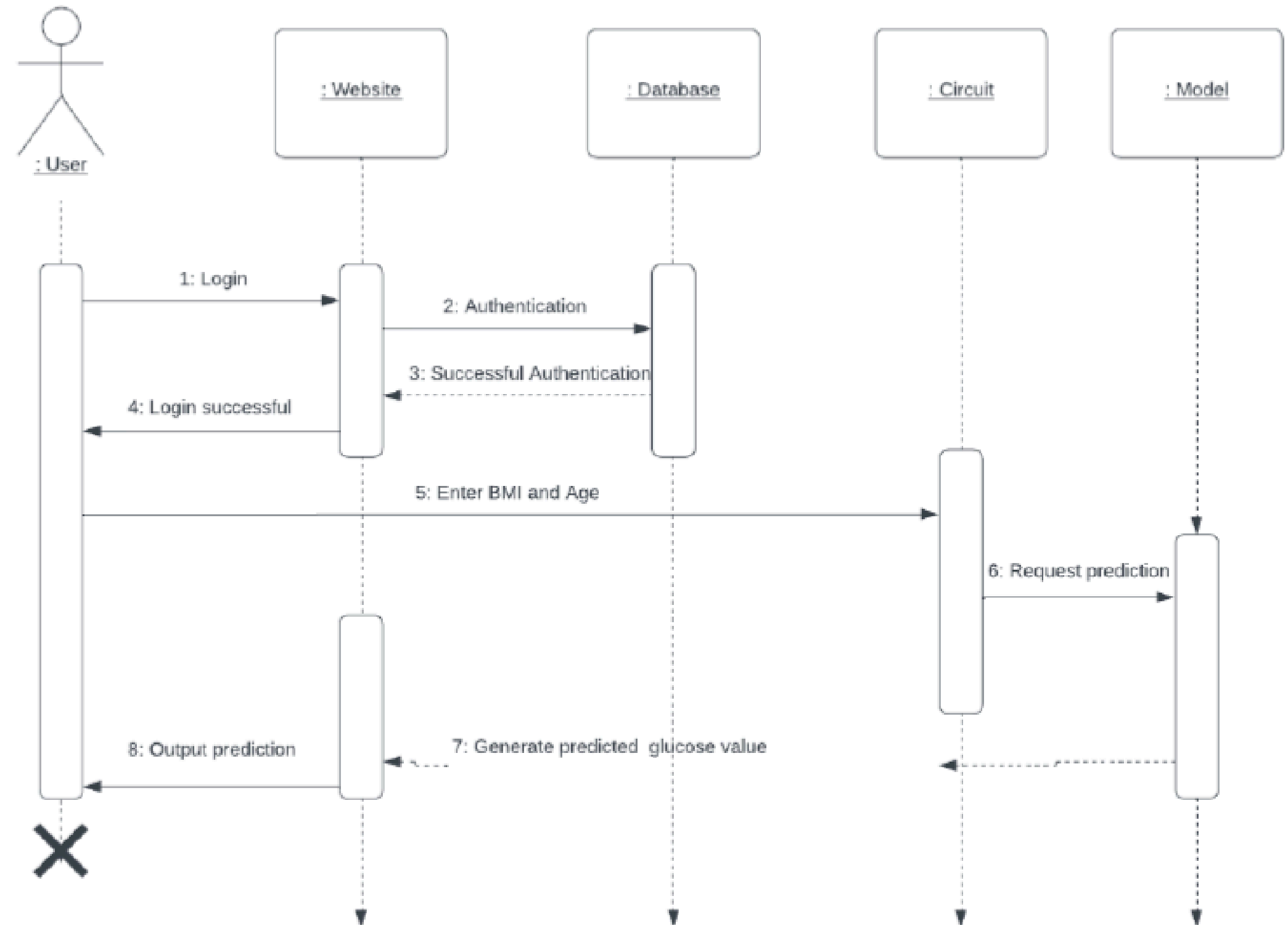
DFD 1



Activity Diagram



Sequence Diagram





Website UI Design

[Website Link](#)

GLUCare

Blogs

About Us

Contact Us

My Profile

LOGIN

GLUCare

An Advanced Non-Invasive Glucose Monitoring System using NIR Spectroscopy.

Sign Up

Name

Email

Age

Weight (kg)

Gender

Password

SIGN UP

My Profile

Name: John Doe

Email: johndoe@gmail.com

Age: 30 years

Gender: Male

Weight: 42 kg

Glucose Tracker

Glucose Value

ADD

Previous Glucose Records

Glucose Value: 120

Recorded on: 12/17/2024, 3:55:02 PM

Contact Us

Name *

Email *

Message *

Login

Email

Password

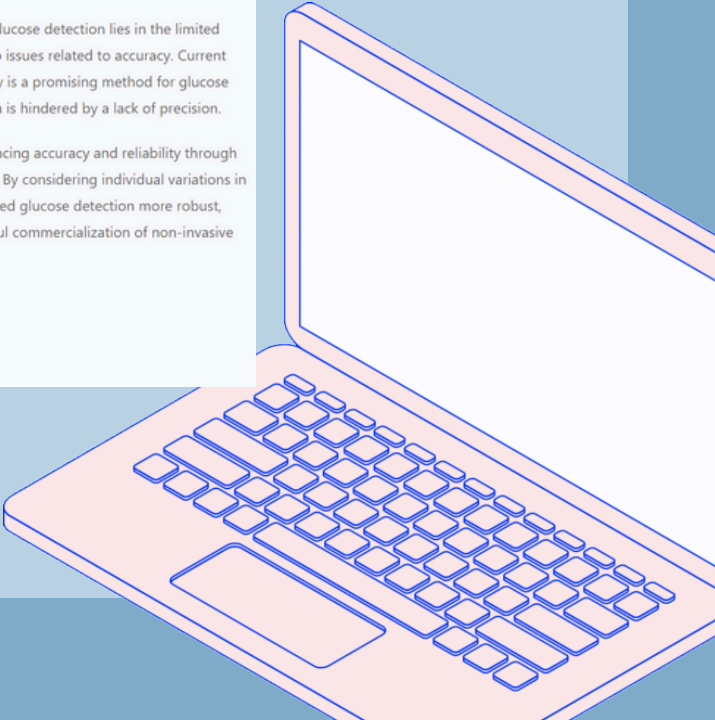
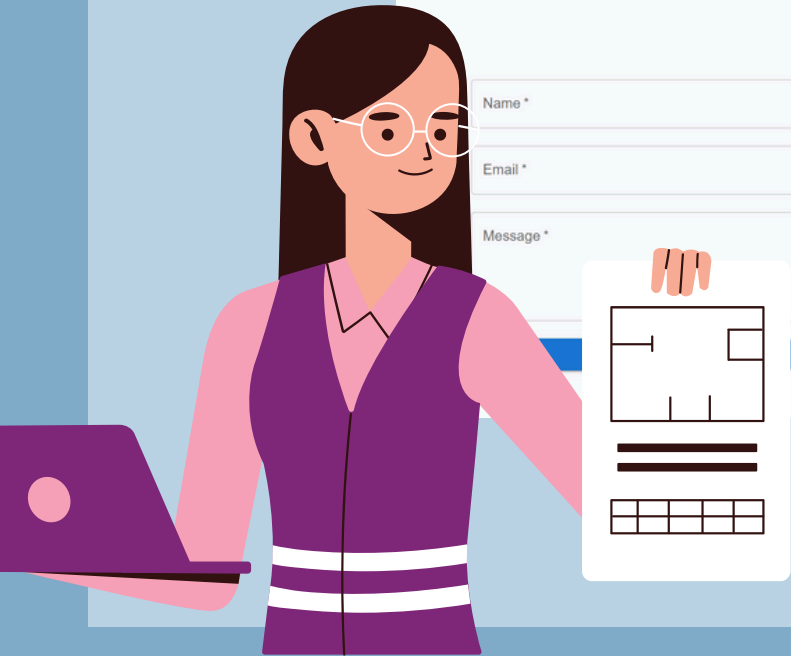
LOGIN

NEW HERE? SIGN UP

About GLUCare

The existing challenge in non-invasive glucose detection lies in the limited commercialization of NIR devices due to issues related to accuracy. Current research suggests that NIR spectroscopy is a promising method for glucose monitoring, but its widespread adoption is hindered by a lack of precision.

Our project addresses this gap by enhancing accuracy and reliability through the incorporation of clinical parameters. By considering individual variations in physiology, our goal is to make NIR-based glucose detection more robust, thereby paving the way for the successful commercialization of non-invasive glucose monitoring solutions.



Project Deliverables

01.

To develop an accurate and reliable non-invasive blood sugar level monitoring device.

02.

Validate the proposed system using current testing methods across different demographics.

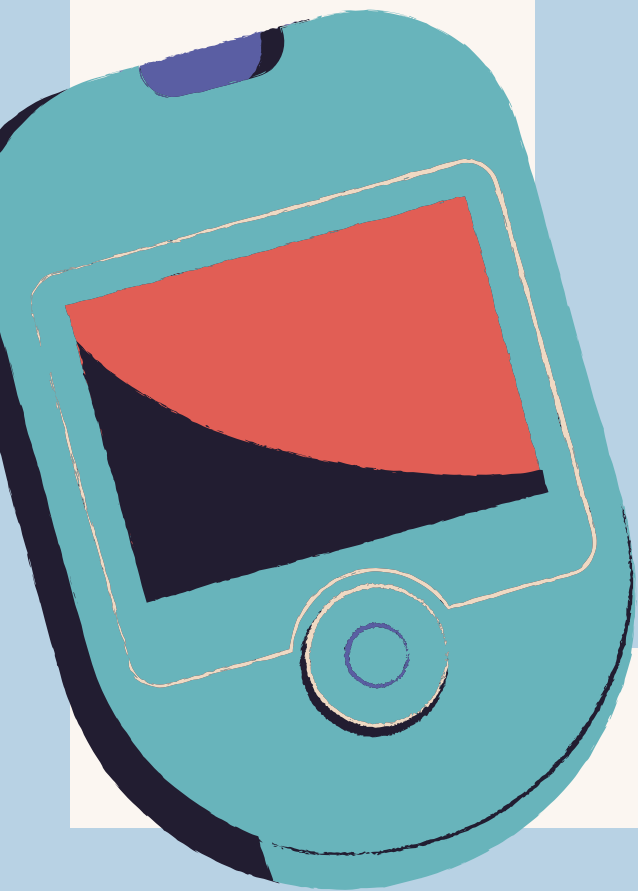
03.

Develop a user friendly web application which will track user's blood sugar level.



Professional and technical learning

1. Gained understanding of Near-Infrared (NIR) Spectroscopy and its application in glucose measurement.
2. Designed and tested algorithms for handling data preprocessing, outlier detection, and feature selection.
3. Utilized hardware integration techniques for connecting sensor data streams to computational models.
4. Improved ability to manage project timelines, testing phases, and iterations for optimization.
5. Gained insights into potential applications of the project, such as continuous glucose monitoring and integration with IoT for remote healthcare solutions.



Role/ Contributions of Individual Team Members

Research Work

Pia Stuti
Anupriya
Nitleen

Design and study of hardware

Nitleen
Pia
Stuti

Model Training

Stuti Riddhi
Nitleen
Pia

Web application developement

Anupriya Pia
Nitleen
Riddhi

Integration of hardware & software

Nitleen Riddhi
Pia Stuti
Anupriya

Testing

Pia Stuti
Anupriya
Nitleen

References

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https://diabetesatlas.org/idfawp/resource-files/2021/07/IDF_Atlas_10th_Edition_2021.pdf

On the use of fiber lasers in non-invasive blood glucose monitoring

<https://www.sciencedirect.com/science/article/pii/S1068520022000050>

Non-Invasive Methods of Glucose Measurement: Current Status and Future Perspectives

https://www.researchgate.net/publication/221892032_Non-Invasive_Methods_of_Glucose_Measurement_Current_Status_and_Future_Perspectives

Wearable-band type visible-near infrared optical biosensor for non-invasive blood glucose monitoring

<https://www.semanticscholar.org/paper/Wearable-band-type-visible-near-infrared-optical-Rachim-Chung/eebf89973bc64fcfdb3dab1220864f918e0810f>



Thank
you very
much!

